Boost Converter Power Stage Calculation (12V to 24V)

Given Parameters

- Input Voltage (V_{in}): 12V
- Output Voltage (V_{out}): 24V
- Output Power (P_{out}): 120W
- Switching Frequency (F_s): 100 kHz
- Ripple Current Rate: 30%
- Ripple Voltage Rate: 1%

1. Duty Cycle (D)

For a boost converter in CCM:

$$D = 1 - \frac{V_{in}}{V_{out}} = 1 - \frac{12}{24} = 0.5 \quad (50\%)$$

2. Output Current (I_{out})

$$I_{out} = \frac{P_{out}}{V_{out}} = \frac{120}{24} = 5\,\text{A}$$

3. Output Resistance (R_{out})

$$R_{out} = \frac{V_{out}^2}{P_{out}} = \frac{24^2}{120} = 4.8\,\Omega$$

4. Inductor Current Calculations

- **Average Inductor Current** ($I_{L_{avg}}$):

$$I_{L_{avg}} = \frac{V_{out} \cdot I_{out}}{V_{in}} = \frac{24 \cdot 5}{12} = 10\,\text{A}$$

- **Ripple Current** (ΔI_L):

$$\Delta I_L = 0.3 \cdot I_{L_{avg}} = 0.3 \cdot 10 = 3\,\text{A (peak-to-peak)}$$

- **Maximum/Minimum Inductor Current**:

$$I_{L_{max}} = 10 + \frac{3}{2} = 11.5\,\text{A}, \quad I_{L_{min}} = 10 - \frac{3}{2} = 8.5\,\text{A}$$

5. Inductor Value (L)

$$L = \frac{V_{in} \cdot D}{F_s \cdot \Delta I_L} = \frac{12 \cdot 0.5}{100 \times 10^3 \cdot 3} = 20\,\mu\text{H}$$

6. Output Capacitor (C)

- **Ripple Voltage (ΔV_{out})**:

$$\Delta V_{out} = 0.01 \cdot V_{out} = 0.24\,\text{V}$$

- **Capacitor Value**:

$$C = \frac{I_{out} \cdot D}{F_s \cdot \Delta V_{out}} = \frac{5 \cdot 0.5}{100 \times 10^3 \cdot 0.24} \approx 104\,\mu\text{F}$$

7. MOSFET Selection

- Voltage Rating: $\geq 24\,\text{V}$ (Choose 30V for margin)
- Current Rating: $\geq 10\,\text{A}$ (Choose 15A for margin)
- Example: **IRF540N** (100V, 33A)

8. Diode Selection

- Reverse Voltage: $\geq 24\,\text{V}$ (Choose 30V Schottky diode)
- Average Current: $\geq 5\,\text{A}$
- Example: **STPS20H100CT** (100V, 20A)